

Ghilas Bettar

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DATA SCIENTIST /AI ENGINEER / ML ENGINEER

PROFILE SUMMARY

Data Scientist and AI/ML Engineer specializing in **Generative AI (GenAI)**, **RAG architectures**, **LLM agents**, and **full-stack MLOps**. Proven expertise delivering **end-to-end ML and Gen AI solutions** across **Healthcare, Finance, and Oil & Gas** domains. Skilled in building and deploying **production-grade AI/ML pipelines** on **AWS (SageMaker, Bedrock, Lambda, Step Functions)** and **Azure ML**, with strong proficiency in **PyTorch, Hugging Face, LangChain**, and vector databases. Adept at implementing **CI/CD automation** (Terraform, CodePipeline, GitHub Actions), containerizing **scalable microservices** (FastAPI/Flask), and driving enterprise adoption of cloud-native AI solutions.

TECHNICAL SKILLS:

AI & Machine Learning: Generative AI (Gen AI), Agentic AI, Large Language Models (GPT, BERT, Transformers), Retrieval-Augmented Generation (RAG), MCP, Agents, Prompt Engineering, Knowledge Graphs, Fine Tuning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Predictive Modeling, Recommendation Systems, Time Series Forecasting (ARIMA, Prophet, RNN/LSTM), Supervised learning & **unsupervised learning**(Classification, Regression, Clustering, K-Means, DBSCAN, Hierarchical), Reinforcement Learning, Feature Engineering, Vector & Semantic Search, Knowledge Graphs, Seq2Seq Models.

MLOps, Deployment, Automation & Monitoring: End-to-end machine learning lifecycle management, Model deployment, Model monitoring & drift detection, CI/CD (GitHub Actions, AWS CodePipeline/CodeBuild, Terraform), Amazon CloudWatch, Prometheus/Grafana, Azure ML, Azure Functions, Databricks, Containerization and orchestration (Docker, Kubernetes), MLflow, Airflow, Feature stores, A/B testing, FastAPI and Flask microservices, Experimentation frameworks, Performance optimization.

Programming Languages: Python, SQL, R, JavaScript, MATLAB.

Frameworks, Libraries & Tools: PyTorch, TensorFlow, Keras, Hugging Face, LangChain, LangGraph, LangSmith, CrewAI, OpenAI SDK, Scikit-learn, PySpark, SparkSQL, NumPy, Pandas, SciPy, NLTK, SpaCy, OpenCV, PIL, Matplotlib, Seaborn, Plotly.

Databases, Vector Stores, ETL and Data Integration: Pinecone, ChromaDB, Faiss, Elasticsearch, OpenSearch, Oracle, PostgreSQL, MySQL, Node.js, Apache Airflow, Azure Data Factory (ADF), AWS Step Functions.

Data Science & Statistical Methods: Exploratory Data Analysis (EDA), Statistical modeling, PCA, Hypothesis testing, Regression analysis, Survival analysis, Feature selection, Hyperparameter tuning, A/B testing.

IDEs & Development Tools: Git, GitHub, Jupyter Notebook, Google Colab, PyCharm, VS Code, RStudio. Cursor, UV.

Data Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly.

Other Skills:

Agile and Scrum: Experienced in iterative development and project management.

PROFESSIONAL EXPERIENCE

MLOps/ Gen AI Engineer

As a Lead Data Scientist/Gen AI at Thermo Fisher Scientific Inc., I spearheaded projects developing AI algorithms for personalized treatment plans and defining customer segments through clustering algorithms, boosting sales by 11%. I optimized AI models using A/B testing to maximize ROI and operational efficiency, while leveraging big data analytics to improve treatment efficacy by analyzing genetic and clinical data. I have also designed an automated Document Summarizer for Multi-Format Inputs through a chatbot.

Responsibilities:

- Integrated the chatbot with LangChain for seamless prompt management and text processing, enabling summarization of plain text, PDF, and Word documents.
- Building data analytics bot to communicate with the database and extracting business insights using charts and plots
- Rolling out fully functional scalable chat application used by thousands of users built using Langchain tools, agents and vector databases
- Enabled handling of large documents by breaking them into manageable chunks, ensuring optimal performance within API token limits.
- Provided users with options to input text directly or upload documents (PDF/Word) for summarization, ensuring a user-friendly experience.
- Proficient in using various generative AI tools and platforms, such as OpenAI's GPT-3, DALL-E 2, and Midjourney.
- Skilled in GEN Ai crafting effective prompts and instructions to generate high-quality text, images, and other creative content.
- Designed and implemented complex agent-based workflows using frameworks like PySyft and Mesa, enabling decentralized and collaborative AI systems.
- Designed Chunking to Embeddings and storing them in VectorDB using PineCone.
- Utilized embeddings to capture semantic meaning of text, enabling advanced document retrieval and similarity search within the summarization system.
- Enabled scalable storage and retrieval of document vectors, allowing for fast and accurate similarity-based searches and enhanced document summarization capabilities using GPT 3.5 Turbo and 4.
- Built a chatbot on patient's healthcare records using pre trained LLM models and integrated RAG pipelines to generate contextual based responses in output.
- Developed and deployed a robust RAG system leveraging PyTorch to the fine-tune large language models (MLLs) on domain-specific data, achieving significant improvements in accuracy and relevance for knowledge-intensive tasks.
- Implemented efficient data pipelines to preprocess and curate diverse datasets, enabling the training of high-quality RAG models capable of handling complex queries and generating informative responses.
- Developed complex analytics pipelines using Apache Spark and Apache Flink for batch processing and real-time data processing from medical devices and sensors.
- Leveraged AWS Glue for serverless ETL tasks, simplifying data transformation processes and reducing operational overhead.
- Proficiently utilized AWS SageMaker for comprehensive machine learning model development, including data preprocessing, feature engineering, model training, deployment, and monitoring with ensuring scalability and high availability.

- Successfully implemented Indico IDP to automate document processing task, such as data extraction, classification, and entity recognition, improving efficiency and accuracy.
- Leveraged Indico IDP's machine learning capabilities to improve data accuracy and reduce manual effort, resulting in significant time and cost savings for the organization.
- Optimized machine learning models on SageMaker, ensuring efficient resource use and cost-effectiveness.
- Designed and managed data workflows using Apache Airflow, creating and maintaining DAGs to automate complex data pipelines.
- Used Spark's RDD and DataFrame APIs for distributed data processing, integrating machine learning libraries such as Apache Mahout and TensorFlow for model training and evaluation.
- Used Git for version control and collaboration on GitHub.
- Built solutions using LLMs (Transformers, BERT, GPT, BART) and designed a penetration attribution model using LGBM for ROI optimization.
- Developed an interactive StreamLit application to run intervention simulations, visualize causal relationship graphs, and communicate insights to stakeholders.
- Created regression models for different products (SKUs) to generate pricing and promotion elasticities.
- Implemented models to predict KPIs, increasing customer retention by 52% through precision-based retention prediction.
- Infrastructure as Code proven experience in leveraging Terraform to automate the provisioning and management of infrastructure on cloud platforms on AWS and Azure, ensuring consistency, reproducibility, and reducing manual effort.
- Capable of collaborating with AI models to enhance creativity and problem-solving abilities.
- Developed several ready-to-use machine learning model templates, assigning clear descriptions of purpose and input variables.
- Developed various machine learning models using Python libraries (Pandas, NumPy, Seaborn, Matplotlib, SciKit-learn) and built/analysed datasets using Python and R.
- Applied linear regression in Python and SAS to understand relationships and causal connections between dataset attributes

ML/MLOps Engineer with FinancialComp

Built and deployed ML-based fraud detection systems end-to-end, covering data engineering, feature engineering, modeling, AWS-based MLOps, CI/CD automation, Dockerized microservices, and production model monitoring.

Responsibilities

- Built distributed preprocessing and feature engineering pipelines using PySpark for large-scale datasets (10M+ records), enabling efficient ML-ready data transformations.
- Performed advanced model tuning using Scikit-Learn's Grid Search and Randomized Search; addressed extreme class imbalance through stratified sampling and optimized resampling strategies.
- Developed anomaly detection models using Local Outlier Factor, Isolation Forest, K-Means, DBSCAN, GMM, and custom NumPy implementations to improve fraud detection coverage.
- Conducted extensive feature analysis and EDA using SciPy, Plotly, and domain-driven insights to support model interpretability and feature selection.
- Automated data ingestion and preprocessing workflows using AWS Step Functions, Lambda, and S3, ensuring consistent, reliable, and scalable data pipelines.

- Designed and deployed end-to-end ML workflows on AWS SageMaker, including training jobs, model packaging, real-time SageMaker Endpoints, and scalable inference pipelines.
- Implemented CI/CD automation with GitHub Actions and AWS CodePipeline/CodeBuild for automated retraining, testing, container builds, and deployment.
- Integrated SageMaker Model Monitor for observability, tracking drift, latency, and KPI changes (AUC, Recall, F1-score) with automated alerts.
- Containerized fraud detection models using Docker and deployed them as scalable FastAPI microservices within the core transaction-processing system.
- Validated production SQL data pipelines, reconciled discrepancies with external systems, and collaborated with SMEs and regulatory teams to refine features and ensure compliance.
- Created dashboards with Tableau and delivered insights through reports, visualizations, and performance summaries for stakeholders and leadership.

Data Scientist at TotalEnergies, Paris, France June 2018 – June 2021

Designed and delivered end-to-end ML solutions to support predictive analytics, forecasting, and operational optimization across the energy value chain. Developed production-grade models on Azure and Databricks, ensuring reliable pipelines, scalable deployments, and user-friendly tools adopted by business teams.

Responsibilities

- Developed predictive maintenance and survival analysis models to reduce equipment failures and optimize maintenance schedules across upstream and downstream operations.
- Built time-series forecasting models (ARIMA, Prophet) to predict power generation and consumption across residential, commercial, and industrial sectors.
- Engineered forecasting models for LNG demand across daily, and monthly horizons, deploying solutions through Azure Machine Learning for centralized access and monitoring.
- Designed and managed ETL and feature engineering pipelines in Azure Databricks using PySpark and SparkSQL to process large-scale sensor, operational, and market datasets.
- Implemented MLOps workflows using Azure ML, MLflow, Docker, and Databricks, enabling experiment tracking, reproducible training, model registry, and automated deployment.
- Containerized ML models with Docker and deployed them as scalable inference services on Azure Kubernetes Service (AKS) for low-latency, high-availability production workloads.
- Integrated ML model outputs into domain applications used by engineers and analysts, ensuring solutions aligned with user skill levels and operational workflows.
- Applied clustering techniques, anomaly detection, and ensemble methods to identify operational inefficiencies and detect unusual events in production assets.
- Collaborated with data engineers, SMEs, and optimization teams to validate data pipelines, incorporate domain expertise, and align models with energy trading and operational objectives.

- Delivered dashboards and insights using Tableau, enabling executives and field operators to evaluate model performance, forecasts, and operational KPIs.
- Documented ML workflows, best practices, and model governance procedures to support Responsible AI guidelines, transparency, and cross-team collaboration.

Data Scientist/ Computer Vision Engineer with Elevance Health, Indianapolis, Indiana Jun 2021 – June 2023

- Designed OCR and NLP models to extract structured clinical information from medical reports, prescriptions, lab summaries, and hospital documentation—reducing manual review time and improving data quality for clinical assessments.
- Leveraged **Azure Cognitive Services (Computer Vision, Form Recognizer, Text Analytics)** to enhance OCR accuracy and accelerate development of document-processing pipelines.
- Built and fine-tuned CNN/RNN-based OCR architectures using TensorFlow and PyTorch, integrating **CTC loss** to improve recognition of variable-length medical text.
- Developed feature extraction, text classification, and NER models to identify diagnoses, treatments, patient demographics, and clinical outcomes from unstructured text.
- Implemented large-scale preprocessing and annotation workflows on **Azure Databricks**, using PySpark to prepare diverse medical datasets for modeling and evaluation.
- Stored and managed large volumes of medical documents and OCR outputs using **Azure Blob Storage**, ensuring secure, scalable, and compliant data access.
- Containerized CV/NLP models with Docker and deployed them as inference services on **Azure Kubernetes Service (AKS)**, enabling scalable, low-latency processing of clinical documents.
- Used **MLflow** within Azure Databricks for experiment tracking, model comparison, and version control, improving reproducibility and collaboration.
- Applied anomaly detection and clustering techniques to identify irregular clinical patterns and support epidemiological analysis tasks.
- Conducted extensive EDA and statistical validation to ensure fairness, transparency, and robustness of AI models deployed in a public health context.
- Collaborated with clinicians, medical experts, and regulatory teams to validate extracted insights, structure clinical rules, and ensure compliance with ethical AI guidelines.
- Developed dashboards and analytics using Power BI to present insights on document processing accuracy, model performance, and population-level medical trends.
- Authored documentation on data workflows, model governance, and AI best practices to support long-term maintainability and responsible deployment of AI systems.

EDUCATION

Master's degree in Operational Research Engineering

University of Bejaia, Bejaia, Algeria

CERTIFICATIONS

AWS Certified: Machine Learning Engineer – Associate

Microsoft Certified: Azure AI Engineer Associate